Based Rollups

Status	In Progress 🔥
Tags	Research
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This is an extended version of Based Rollups in Rollup Types

What is a based rollup?
- Existing rollup sequencing architecture
- Benefits of a based rollup
- Design options
- What do we need for Based Rollups to happen?

The road to (better) based rollups
- Based Preconfirmations
  - What are preconfirmations?
  - How?
  - With/out Execution Tickets
  - Restaking Relevance: EigenLayer and Karak
- Proving
- TEEs
- Inclusion Lists
  - FOCIL

Based Rollup Landscape
- Gwyneth

Question box

Resources

What is a based rollup?

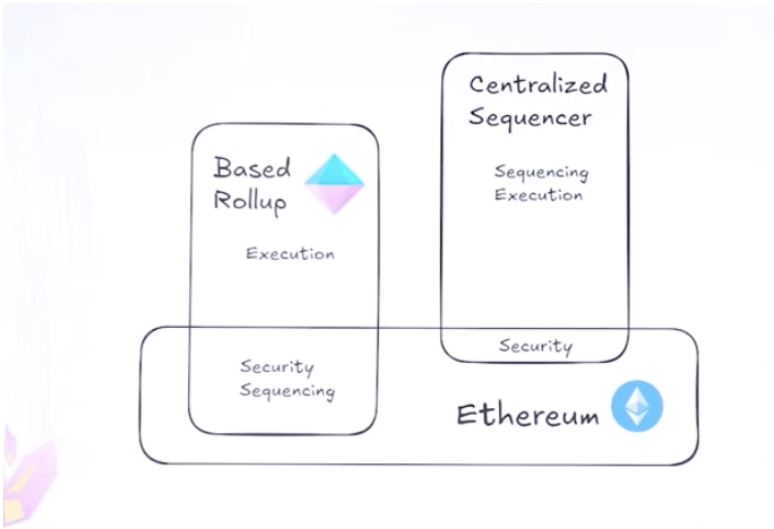
Rollups hold three different aspects: DA, consensus and execution.

Based Rollups use `base sequencing` as the method for sequencing transactions and block building. With this method, L1 `consensus` (proposers, validators) holds the role to decide the ordering of transactions of the rollup.

> *"More concretely, a based rollup is one where the next L1 proposer may, in collaboration with L1 searchers and builders, permissionlessly include the next rollup block as part of the next L1 block."*

Consensus of the L2 remains tied to L1 while execution remains outside of Ethereum L1 and other DA methods can be used. All of this without any change needed to L1. I fact we already have two based rollups in mainnet: Taiko and Facet.

Additionally, in based rollups, data compression would happen offchain, as in optimistic rollups, and then posted to L1. If this were a ZK rollup, builders would also have to make ZKPs.



Credit to Spire Labs

To compare with other rollup architecture:

	Optimistic	ZK	Based
Proofs	Fraud proofs	ZK proofs	Fraud Proofs
Sequencer	Rollup	Rollup	L1
Consensus	L1	L1 w/ own method of ZK proof validation	L1
Execution	Rollup	Rollup	Rollup

Existing rollup sequencing architecture

A `centralized sequencer` is a point of failure for a rollup as sequencing requires us to rely on a trusted entity.

While centralized sequencers are easier to launch, protect us against MEV and allow for quicker transactions (as well as low costs!), they have different drawbacks:

- leverage trust of entities (OP, Arbitrum)
- cause fragmentation issues and are intra- not interoperable (e.g. Optimism's interoperability remains within Superchain chains).
- L2s with centralized sequencers can "compete with L1", MEV can be captured by sequencers on L2.
- *not* censorship resistant: sequencers may indefinitely censor transactions

- are central point of failure
  - ▼ On liveness in current L2s

    On a non-based rollup, if a centralized sequencer goes offline or becomes unavailable, the recovery time can be too long and transaction finality can be delayed.

    In such cases users must resort to L1 as an escape hatch.

- *not* credible neutral

With `PBS` via MEV-Boost the sequencing rights are given to `consensus participants`, builders and searchers do the heavy lifting. Whoever wins the MEV Boost auction, wins the right to order both L1 and based rollup transactions. In the future the L1 will change how it sequences their transaction with proposals like FOCIL ePBS and more.

PBS and ePBS

Benefits of a based rollup

- L1 is the most **secure sequencing** and enhances **decentralization** in the sequencing of transactions
- Rollups can have seamless **access to L1 state and permissionless inclusion on L1 blocks.**
- **Increased Liveness**: In a based rollup, if a sequencer fails users only need to wait for the next block proposer for their transaction to be included.
- no need to keep up with L1 upgrades which are currently done through governance.
- Kinda solves the extra costs in the appchain thesis. Apps can take advantage of infrastructure that already exists in L1 (like oracles), which can be costly for medium to small apps if they were to launch a chain. You can have maximum customization of environment.
- no gas overhead linked to traditional rollup models, based rollups bypass the required token-based operations. There is no need to verify L2 signatures – it's all done by L1.
- In case of shutdown rollup native assets don't go to zero (memecoins, NFTs)
- economic alignment with L1, as MEV flows back to L1 and social alignement (instead of leveraging L2 trust). We scale maintaining L1 network effects

Using consensus for sequencing unlocks `synchronous composability,` as if you had a shared sequencer. With synchronous composability contracts can call any other contract at will on a single block and interact with other based rollups, and enjoy

shared liquidity across chains while retaining low cost transactions.

👍 less fragmentation

👍 seamless ux for users

👍 seamless developer experience

👎 12 second `soft confirmation` times (dictated by L1 block/slot times)

▼ `What are soft confirmations?`

Soft confirmations refer to the ability for a rollup user to reliably know that their transaction will successfully reach the L1.

When the rollup commits by posting their ordered transactions you achieve soft confirmation, promising to include transactions.

Hard confirmations are achieved when inclusion is confirmed by L1, after 12 seconds.

Design options

▼ What are `Subslot Auctions`?

Subslot auctions sell access to state for under 12 seconds block time.

Auctions are carried out as a way to capture MEV. Sequencing rights can also be distributed by different non-auction methods e.g. donating to a solo staker or to a public good to optimize other values.

Independent subslot auctions are not dependent to each other within a block level.

- `Total Anarchy`: There is no rule to define who the sequencer is. You can't do preconfirmations. MEV leakage to L1. It's the current Taiko model and the original based sequencing model proposed by Justin Drake.
- `Vanilla Based Sequencing`: Adds sequencing election beforehand with random selection of proposers
  - MEV Leakage still happens but includes preconfirmation. Sequencing revenue beyond the gas thats being charged (all forms of MEV) is captured by L1 proposer.
  - MEV capturing cant be measured
- `Complex Sequencer Election Mechanisms`: Don't always delegate to L1, building an in-between layer with an auction mechanism to win sequencing rights.
  - Gwyneth and Spire approach: Sell tickets in a permissionless auction to anyone, but actually using them can only be used by a L1 proposer. If the winning bid is not a proposer, they have to sell them to someone who can use them.
  - Espresso proposes to run an auction, giving the L1 proposer a last look so that they can "get in" buying at the last price or leave it to the winning bid (right of first refusal).
    - Would now know MEV and revenue generated

What do we need for Based Rollups to happen?

- Reducing slot times → `Preconfirmations`
  - Preconfirmation for transaction inclusion and state

    - Can take the form of collateral based preconfs, L1 puts collateral if you don't comply is lost $\longrightarrow$ can achieve 100 ms slot times.
- `Real time proving`:
    - accelerating TEE development as a way to generate fast zk proofs
    - achieve fast validity proving aggregating validity proofs on many L2s allow composability with l2 or and ethereum
- In the `DA` aspect
    - Data cost reduction on L1 is key for base rollups.
    - But also extending bandwidth, as with current throughput in L1 it would not be able to handle Giga gas (66x of Base's performance).
- Maybe
    - `Attester Proposer Separation (APS)` as a validator you produce inclusion lists and validate the correctness of the chain.
    - `No delay forced inclusion` might be a good idea to become more censorship resistant. It synchronizes the forced inclusion and the default sequencing. If L1 reorgs, BR does too.
    - `Multi concurrent proposers` can increase the number of leaders one can run the auction over. Increasing efficiency and amount of participants, making it likely that a preconfirmation will be done by a better node.

The road to (better) based rollups

How can we reduce confirmation times maintaining assurances *without* introducing centralization?

Based Preconfirmations

What are preconfirmations?

Based preconfirmations are a bottle neck in based rollup development. Without them, transactions will have confirmation times of 12 seconds.

`Preconfirmation` are commitments from validators to users guaranteeing that the transaction will be accepted. Preconf design usually requires a way to align incentives, making that promise subject to slashing. We can then call them collateral based preconfirmations. Preconfirmations can also mean the commitment to the actual state of the L1. So...

👍 faster confirmation times

💫 harder assurances

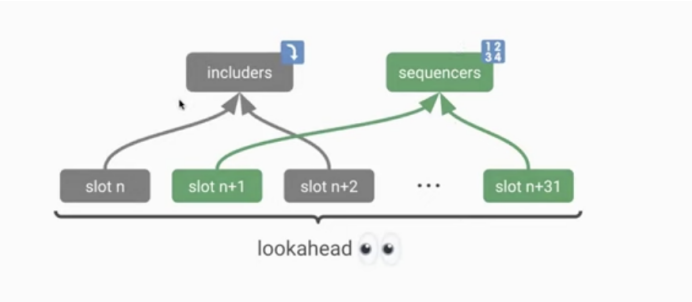
😉 If not designed well, could lead to limited availability of confirmations, slower times and, therefore, limited composability with other chains.

Preconfs are still in design status, with or without execution ttickets and among different teams. There is no established design yet. How section enters into debatable models to preconfirm transaction and state.

How?

Preconfirmations can be achieved with the "lookahead" function. `Sequencer` proposers have full control of the ordering from the head till their end of slot. You can send a RPC request to the next sequencer proposer and request inclusion, if a promise is sent back, then the transaction of the users can be included by the next `includer` proposer.

That confirmation will then be gossiped around



Sequencing Calls #1

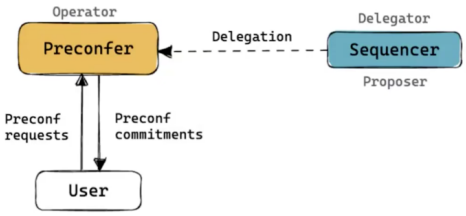
Collateral can be committed by the proposer to align sequencer proposers. Ok, but what if the preconfirmer (sequencer) goes offline? provide a proof that there was no missed

Sequencer proposers can confirm up until 30M gas, so as to not “over confirm” what can fit inside a slot.

With/out Execution Tickets

With `execution tickets` we can make sure that proposers are not validators. Only validators that can run a full node for every rollup, have high bandwidth (10k+ TPS), high uptime and denial of service protection.

Without execution tickets, Justin Drake introduces the notion of a `preconfers node` which would take registrations ahead of time from proposers, delegating their confirmations to these nodes. These are trusted to be online *always*. Waht we need for this is a way to express delegation preferences



Preconfers can be the relays, since they are already a highly sophisticated entity and are already trusted. This would require restaking so as they can be held accountable and slashed if needed. This option also further centralizes power in the relays. Preconfers can also be the own nodes, eliminating the need on restaking.

First we need adoption of validators participating in preconfirmation protocols. For each epoch (32 block slots) we need enough validators participating so that at least one validator participates in that epoch and based transactions can be included.

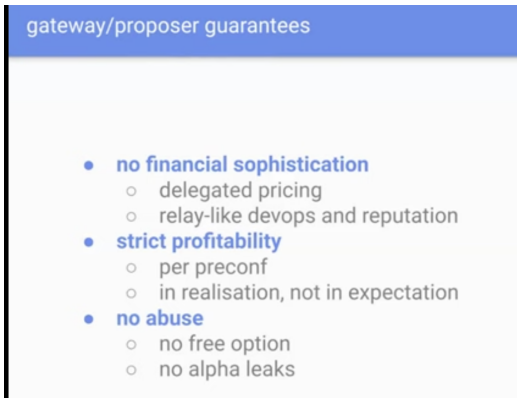
Another way to achieve faster confirmations would be to lower L1 block times to 2 secs, but this would put pressure on the bandwidth needed for validators to keep up with the head of the chain. `Verkle` and `Max EB (Pectra)` would help as they would make L1 more resilient and would reduce blob cost.

> Main goal: Is it possible that the L1 proposer can give me a confirmation for L2 transactions?

Yes, implementing `gateways` (registry contract). Gateways control the sequencing for rollups and control preconfirmation prices.

L1 proposers register by staking a specific amount. They can give you a preconfirmation for the block. Not all proposers will be part of the registry contract so some delay is expected for inclusion, but there is the highest inclusion guarantees.

https://x.com/gwyneth_taiko/status/1878873539144724984



Preconfirmation & ePBS

"ePBS" should make preconfirmations easier to implement than with MEV-Boost. But this is more far out. EIP 7732 illustrates better, more robust designs than MEV-Boost.

Restaking Relevance: EigenLayer and Karak

Restaking becomes relevant in the based rollup talk when talking about preconfirmations. AVS are a way to bootstrap networks of validators for a specific task.

Teams like Nethermind are building preconfirmation AVS for a network of nodes to provide faster assurances that are subject to slashing by the restaking protocols (EigenLayer, Karak, etc.). Taiko plans to adopt Nethermind's AVS. Puffer's Unify will also use a Preconfirmation Slasher AVS, using their own validators, since they already run a Liquid Restaking Protocol with multiple validators in EigenLayer.

With maxeb this can work to emit preconfs.

Proving

When proposing a block, based rollups require to wait for the L2 block to be `proved`.

With the current proving, each block in L1 would include the `SNARK proof` for the previous block, so no new block can be created until the proof is generated. SNARK proof generation takes several minutes, meaning: no guarantee that a user transaction is complete for *minutes. Slower proving times, means its is currently an inefficient way to bring proving L2 blocks.*

Real time proving or near real time proving is on its way.

The things that are available to achieve real time proving are TEEs and AVS solutions to run a proof to verify the state of the blockchains. Right now they are not fully available but will be in 2025.

TEEs

An environment with a secure enclave on a chip (usually Intel chip). Secured by true obscurity, you don't know anything about the hardware when looking at it. Design and implementation is quite obscure too.

Note: Close source hardware means you have to trust manufacturers. Which is the main critique on TEEs: trust assumptions.

To securely use TEEs to prove a block, a mechanism has to be established to be able to contest that proof of the block. You submit a challenge, which if correct then causes the proposer to be slashed.

Inclusion Lists

Full inclusion guarantees for all offchain computations.

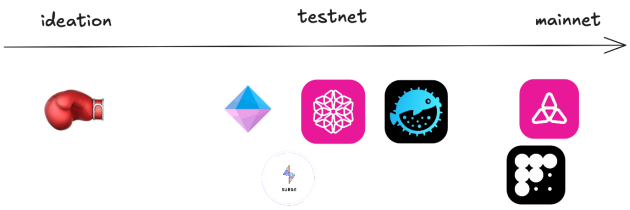
FOCIL

Fork Choice Enforced Inclusion Lists is a type of inclusion list design that is dictated by committees with 16 generators. Proposers must follow the 16 inclusion lists generated.

sidecar software on docker container to become a `pre-confer node`. Lookahead of consensus gives us info of epoch and epoch +1. You choose the one thats a pre-confer node. You can still push

Based Rollup Landscape

aka teams building BR, actual preconfirmation and TEE landscape is bigger.



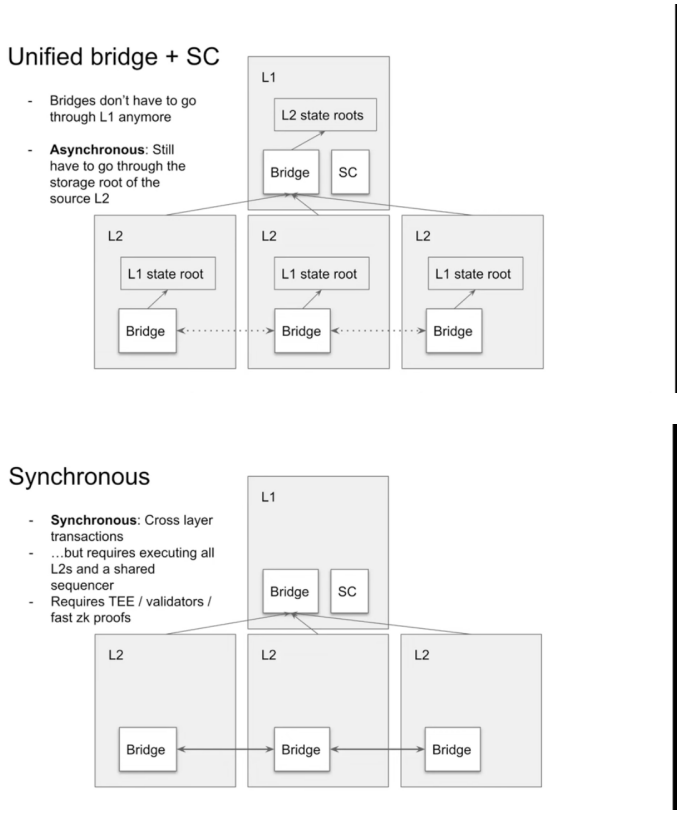
Jan. 2025

	Status	Is Fork?	Sequencer & Percents	Proving system	Block times	MEV leakage/ETs	Problems
`Taiko`	Mainnet	No	Proposer not elected beforehand but it can be ≠ of proposer No preconfs rn. You wait for L1 inclusion. But looking to bring them on mainnet with Nethermind AVS.	None rn, L1 inclusion. A future version will integrate Nethermind AVS accelerating confirmation and block times.	12-33 seconds (slower than Ethereum)	Yes. Execution tickets ***not*** used for MEV price oracle in a block.	– No sync composabili - Can lead t centralizatio around preconfs
`Facet`	Mainnet	Optimism fork	Inclusion guaranteed by a similar to L1 forced inclusion. No preconfs.		12 seconds.		– Expensive – Is Sovereig
`Puffer Unify`	In Testnet and Devnet w/ preconfs. Public testnet soon	No	Puffer validators sequence. Preconfirmations with Puffer Slasher AVS		100 ms	Yes. AVS co-built with Justin Drake	
`Surge`	Development	Taiko fork by Nethermind	Preconfirmations with Nethermind AVS		tbd		
`Spire`	In testnet	No	Complex Sequencer Election based sequencing Based preconfirmations		tbd	Yes similar to ets burn tonpropose a block.	
`Gwyneth Taiko`	Development	No. Boosted rollup Incubated by taiko.	Complex Sequencer Election based sequencing	Multi realtime prover (sub 12 seconds proving). Uses	< L1	Yes	

	Status	Is Fork?	Sequencer & Percents	Proving system	Block times	MEV leakage/ETs	Problems
			Based preconfirmations	Multi types of proofs.			
`Rise`	Internal testnet is live. Public testnet Q1.	No	TBD Based preconfirmations		< L1		
`Rogue`	Ideation: design & research	No	TBD Based preconfirmations	STARKs with a KZG wrapper	< L1	TBD	–

▼ Gwyneth

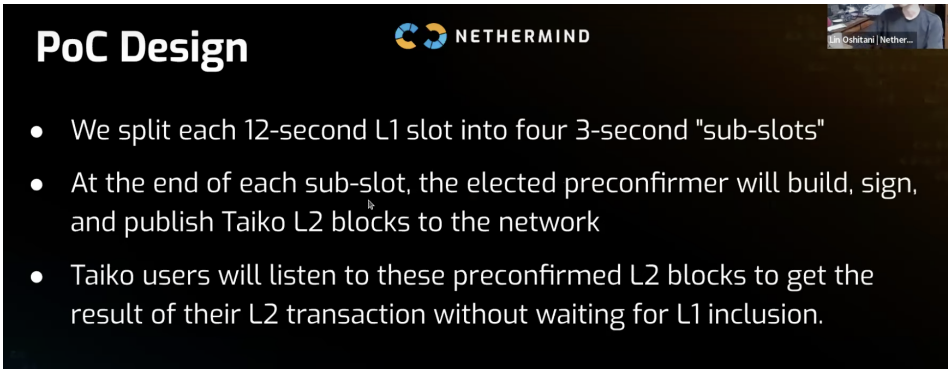
Shared bridge for L2s. You don't touch the L1 but you emit a proof. It's still asyncronous, still has to fo through the storage root of source L2, so it depends on origin L2 finalization times.

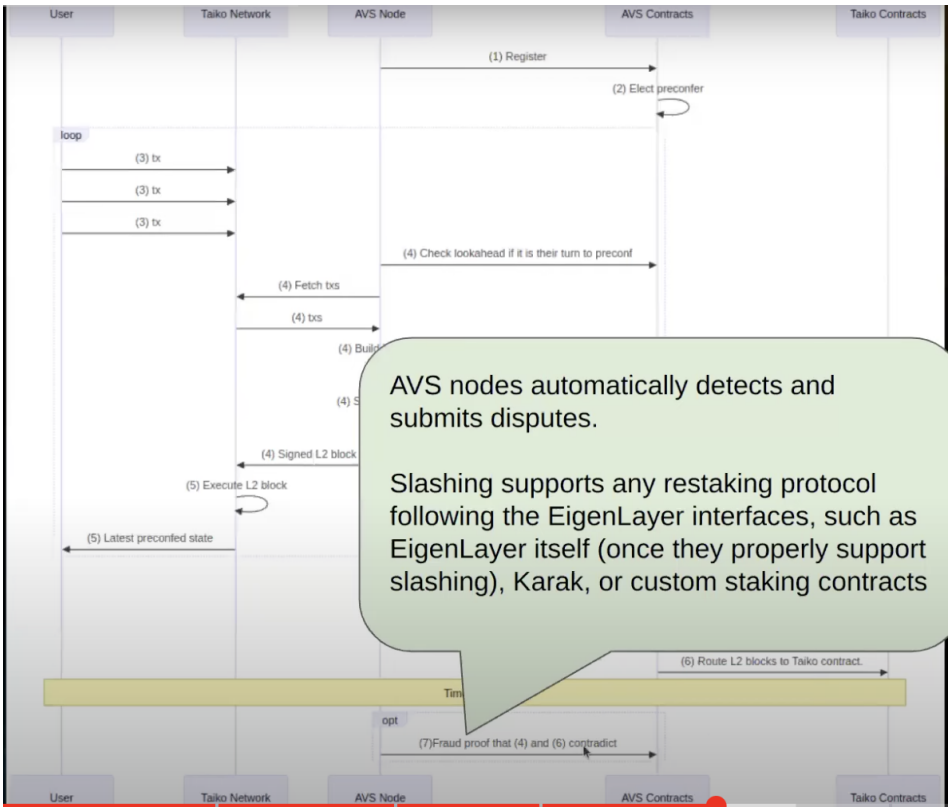


Full node needed to check

validity proof when block is proposed. Requires a TEE.

▼ Nethermind





- fast validity proving aggregating validity proofs on many L2s
- succinct risc 0

💡

Question box

- private mempools - 1.30 / where do txs GO?
- gateways
  - Gateways in preconfs how do they work? validators give preconfs, bandwidth cost and resources cost. → nethermind talk
- Tickets? burn ticket distributed offchain to propose a block.
- Optimizing DA in Ethereum – what is needed?
- Is the role of eigenlayer similar to preconf nodes? or is it just like Puffer validators?
- What do wallets need to consume preconfs?
- facundo articulo encontrar

Add no delay forced inclusion FOCIL

https://x.com/primev_xyz

https://x.com/interstatefdn

Resources

Calls

https://www.youtube.com/watch?v=CJ1otXtBrqA

Resumen calls https://docs.google.com/document/d/1FG3nKQdUNb_YHCp_IzSDkC_r7A6HOT11O2YNUjCX-6s/edit?tab=t.0

Research

https://ethresear.ch/t/based-rollups-superpowers-from-l1-sequencing/15016

Talks

https://www.youtube.com/watch?v=WiKPlNGrUzU

https://www.youtube.com/watch?v=UfoG4cRk9Z0

https://www.youtube.com/watch?v=uI2KSEXhtZc

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Articles + edu

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https://unifi.puffer.fi/files/Puffer-UniFi-Litepaper.pdf

https://www.surge.wtf/

To check

https://www.chaincatcher.com/en/article/2153919

https://gist.ly/youtube-summarizer/unleashing-the-future-of-ethereum-rise-of-base-rollups-and-giga-gas-era

https://taiko.mirror.xyz/7dfMydX1FqEx9_sOvhRt3V8hJksKSlWjzhCVu7FyMZU

https://ethresear.ch/t/understanding-based-rollups-pga-challenges-total-anarchy-and-potential-solutions/21320

https://ethresear.ch/t/based-preconfirmations/17353

related

https://www.youtube.com/watch?v=esGAFXA7j0k&embeds_referring_euri=https%3A%2F%2Fwww.youtube-nocookie.com%2Fembed%2FesGAFXA7j0k

https://mirror.xyz/privacy-scaling-explorations.eth/I5BzurX-T6slFaPbA4i3hVrO7U2VkBR45eO-N3CSnSg

inclusion julian ma

https://ethresear.ch/t/uncrowdable-inclusion-lists-the-tension-between-chain-neutrality-preconfirmations-and-proposer-commitments/19372